

Identifying biologically interpretable features in metagenomic foundation models for DNA synthesis screening

Mannat V. Jain

INTRODUCTION: Screening

How does DNA synthesis screening work?

- Current DNA synthesis screening approaches rely on sequence similarity to known hazards [1].
- Generative AI tools can be used to design synthetic homologs that bypass screening [2].
- Previous work proposed using foundation models for function-based screening by identifying sequences indicating harmful functions [3].

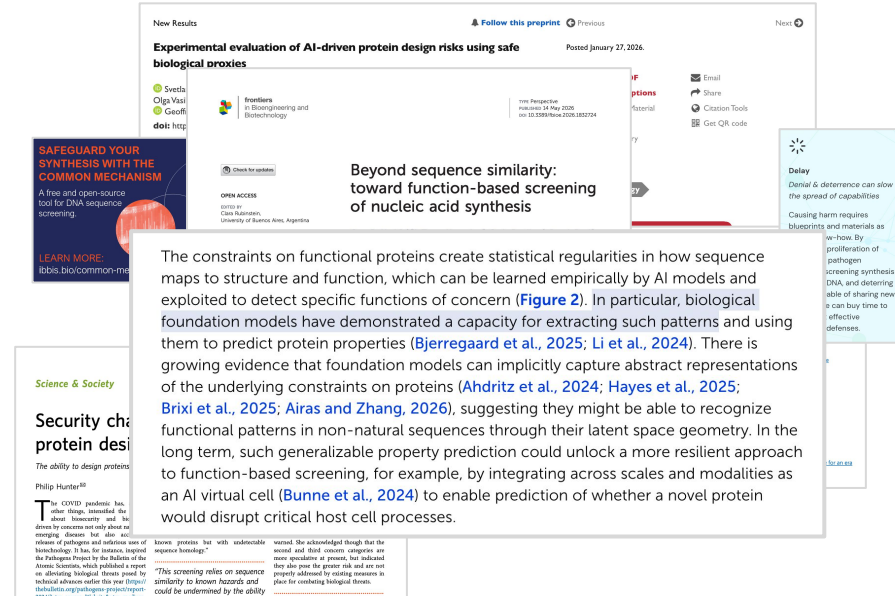


Fig. 1. SELECTED WORK ON AI BIOSECURITY RISK: [1-4], with screenshots from IBBIS and SecureBio.

INTRODUCTION: METAGENE-1

Why METAGENE-1?

- Llama 2-7B architecture trained on 1.69t base pairs of an unreleased dataset collected by SecureBio from sequenced DNA and RNA from wastewater samples.
- Outperformed DNABERT-2, DNABERT-S, and NT-2.5b-Multi on Pathogen-DETECT.
- Byte-Pair Encoding on a 2b subset of the pretraining dataset.
- Evaluate on Gene-MTEB to assess ability to create high-quality zero-shot representations for genomic sequences.

METAGENE-1: Metagenomic Foundation Model for Pandemic Monitoring

Ollie Liu¹, Sami Jaghouar², Johannes Hagemann², Shangshang Wang¹,
Jason Wiemels¹, Jeff Kaufman³, and Willie Neiswanger¹

¹University of Southern California, ²Prime Intellect, ³Nucleic Acid Observatory

We pretrain METAGENE-1, a 7-billion-parameter autoregressive transformer model, which we refer to as a *metagenomic foundation model*, on a novel corpus of diverse metagenomic DNA and RNA sequences comprising over 1.5 trillion base pairs. This dataset is sourced from a large collection of human wastewater samples, processed and sequenced using deep metagenomic (next-generation) sequencing methods. Unlike genomic models that focus on individual genomes or curated sets of specific species, the aim of METAGENE-1 is to capture the full distribution of genomic information present within this wastewater, to aid in tasks relevant to pandemic monitoring and pathogen detection. We carry out byte-pair encoding (BPE) tokenization on our dataset, tailored for metagenomic sequences, and then pretrain our model. In this paper, we first detail the pretraining dataset, tokenization strategy, and model architecture, highlighting the considerations and design choices that enable the effective modeling of metagenomic data. We then show results of pretraining this model on our metagenomic dataset, providing details about our losses, system metrics, and training stability over the course of pretraining. Finally, we demonstrate the performance of METAGENE-1, which achieves state-of-the-art results on a set of genomic benchmarks and new evaluations focused on human-pathogen detection and genomic sequence embedding, showcasing its potential for public health applications in pandemic monitoring, biosurveillance, and early detection of emerging health threats.

🌐 Website: metagene.ai
📦 Model Weights: huggingface.co/metagene-ai
🔗 Code Repository: github.com/metagene-ai

Fig. 2. METAGENE-1: 7-billion-parameter metagenomic foundation model pretrained on wastewater DNA sequences to aid pathogen detection [5].

OBJECTIVES

Does METAGENE-1 learn cross-family biologically functional features not tracked by k-mer composition?

- I. Probe METAGENE-1 for pathogenicity and family, and compare the results to k-mer baselines.
- II. Test whether METAGENE develops organism-specific representations beyond composition.
- III. Identify latents that fire for functional genes, and show they generalize to a held-out viral family.

METHODOLOGY: Data Filtering

How did we obtain and label our dataset?

- 80,000 reads (4 classes x 20,000 reads / class) publicly available in the evaluation dataset for METAGENE.
- 19,645 reads (one per cluster) after eliminating near-duplicates using VSEARCH clustering at 0.95.
- 19,643 reads after eliminating identical duplicates.
- Used VSEARCH to align pathogen reads with HumanVirusReference genomes to identify family.
- Downloaded 18 annotated RefSeq genomes to match reads to genes using blastn.
- Split 80/10/10; also held-out Picornaviridae as proxy for novel viral family.

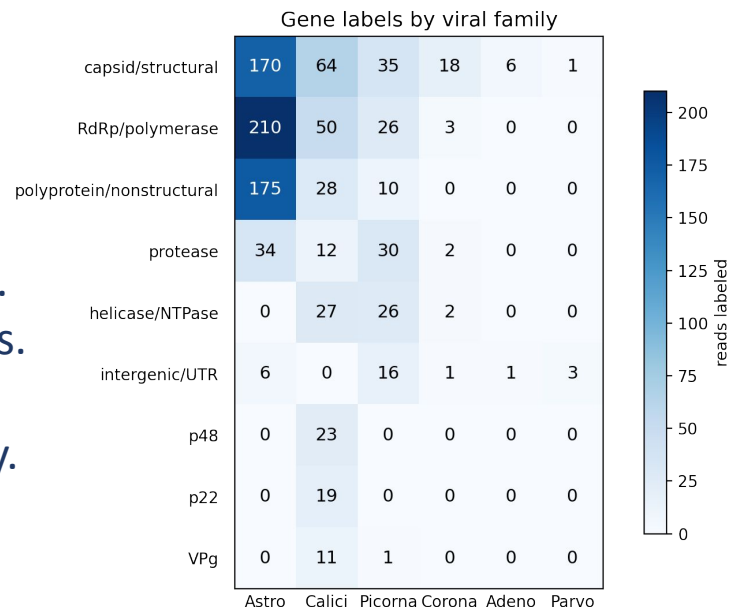


Fig. 3. GENE AND FAMILY LABELS. Used VSEARCH and BLASTn to label reads according to gene and family.

METHODOLOGY: Linear Probes

Does METAGENE linearly encode (a) pathogenicity (b) organism families?

- Linear probes are classifiers trained to predict labels directly from model activations.
- Trained logistic classifiers on mean-pooled residuals after L0, L4, L8, L12, L16, L20, L24, and L31:
 - Binary probe for human-infecting virus / background.
 - Multiclass probe for organism family on labeled reads.
- Evaluated against 6-mer baseline and k-mer ladder.

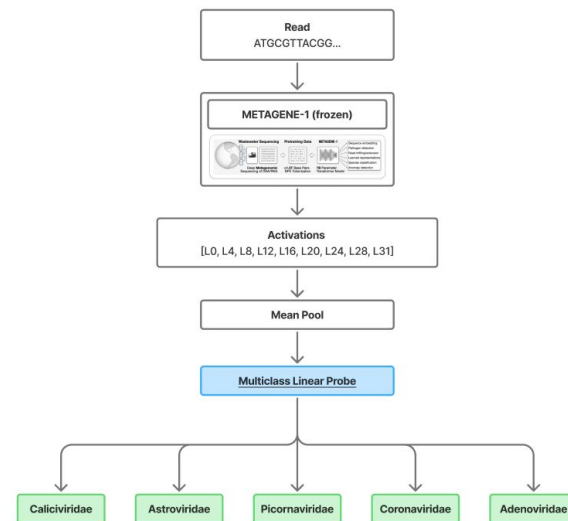


Fig. 4. LINEAR PROBING TO CLASSIFY FAMILY. Trained probes to classify labeled reads as Caliciviridae / Astroviridae / Picornaviridae / Coronaviridae / Adenoviridae.

METHODOLOGY: Sparse Autoencoders

Do SAEs uncover interpretable features in METAGENE's representations?

- Sparse Autoencoders reconstruct model activations in a sparse higher-dimensional space.

$$z = \text{TopK}(W_{\text{enc}}(x - b_{\text{pre}}))$$

$$\hat{x} = W_{\text{dec}} z + b_{\text{pre}}$$

- Trained Top-64 SAEs [7] with sae_lens on token-level activations following recent work on ESM-2 [8].
- L20/L24 SAEs reconstruct 82-85% of variance (FVU 0.15-0.18) with <0.1% dead latents.

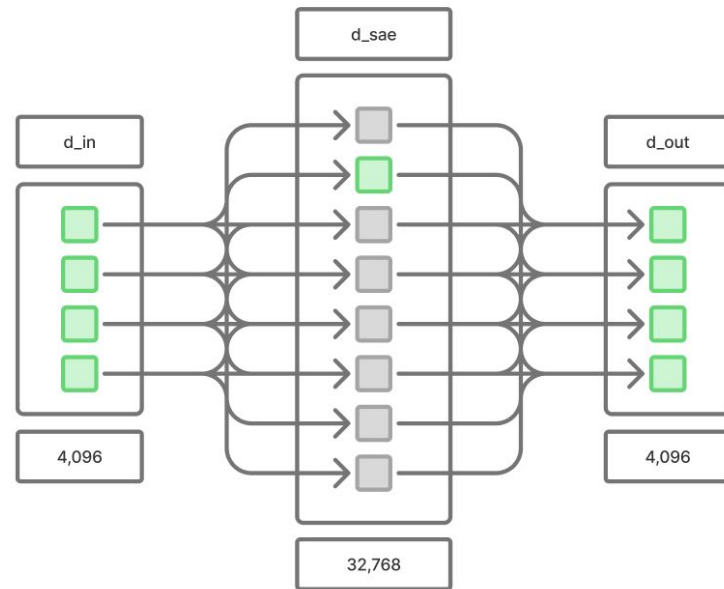


Fig. 5. SPARSE AUTOENCODER DIAGRAM. Top-64 SAEs to reconstruct the activations of 4,096d residual stream in a 32,768d hidden space.

METHODOLOGY: Autointerpretability

Can we identify human-interpretable labels for our latents en-masse?

- Standard autointerpretability [9] is difficult for biology since our input strings are gene sequences.
- Named latents on a FIT dataset (majority gene of top-10% firing reads) and scored on an EVAL dataset using Pearson correlation between firing strength and gene label.
 - Shuffled gene labels within each family ~200 times to isolate gene from family tracking.
 - Tested latent firing on held-out Picorna family reads to test generalization.
 - Trained a k-mer classifier to test whether the composition explains firing.

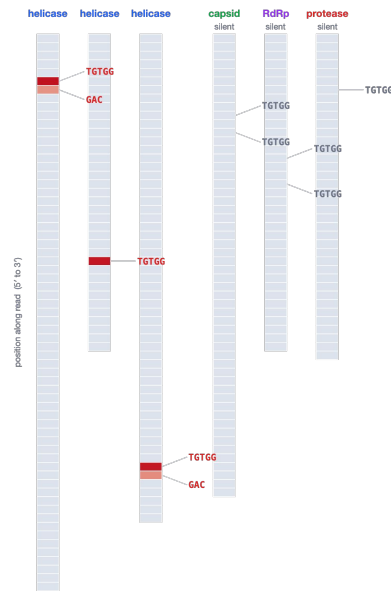


Fig. 6. LATENT FIRING ON READS . Latent firing for TGTGG and GAC in helicase but not on TGTGG in other genes.

RESULTS: Pathogenicity

METAGENE linearly encodes both pathogenicity and organism families.

- Linear probe trained to classify reads as Human-Virus Infecting / Background peaked at L31 with 0.965 AUROC.
- AUROC increases from 0.891 at L0 to 0.965 at L31; this suggests METAGENE is learning relevant biological features.
- Multiclass family probe achieves 0.927 AUROC by L4 and remains ~flat afterwards.
- Pathogenicity is tied by a logistic regression model trained on 10-mer statistics (AUROC 0.966).

Layer	Pathogenicity	Organism family
0	0.891	0.904
4	0.931	0.927
8	0.933	0.920
12	0.921	0.908
16	0.924	0.923
20	0.918	0.927
24	0.930	0.926
31	0.965	0.911

Table 1. LINEAR PROBE PERFORMANCE. Linear probes classifying pathogenicity peaked at L31 at 0.965 AUROC and family at L20 with 0.927 AUROC.

RESULTS: Organism Representations

METAGENE develops emergent representations of organisms.

- Viral families are linearly decodable with AUROC ≈ 0.93 at L24.
- Organism representations are mostly composition [10]:
 - Trained a linear regression model to reconstruct L24 activations from 6-mers.
 - Subtracting the reconstructions dropped L24 family probe from 0.93 to 0.53 AUROC.
- ≈ 200 partial astrovirus latents (best single 0.63 AUROC vs. probe 0.93).

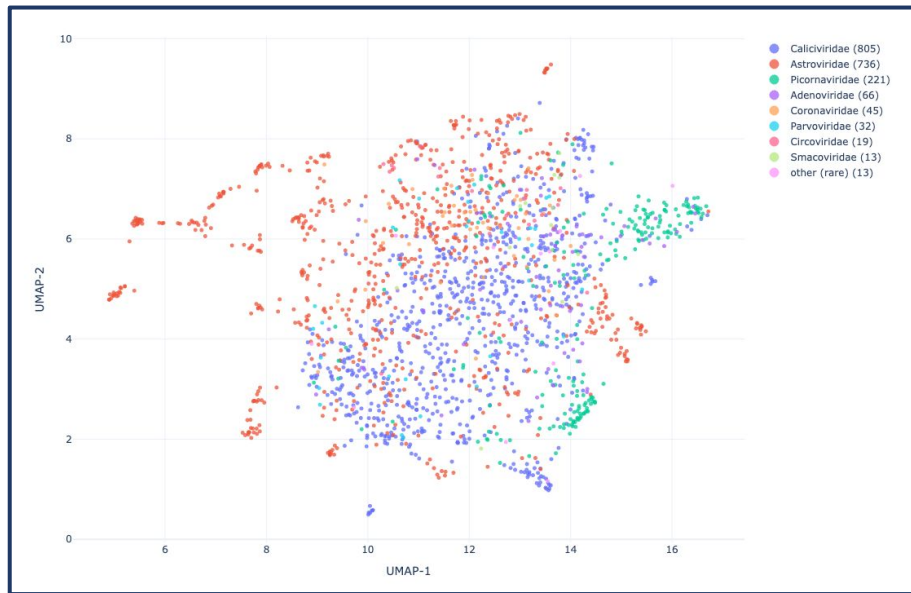


Fig. 7. ORGANISMS CLUSTER. UMAP of mean-pooled L24 activations shows that viral families cluster (kNN 0.448 vs. 0.167 shuffled null).

RESULTS: Functional Latents

METAGENE learns to represent gene function independent of organism.

- METAGENE-1 non-compositionally represents gene function across families.
- Scored named latents by Pearson test (firing strength, gene label) on held-out reads ($r \approx 0.43$, $z = 4.4$).
- Showed 2-mer composition transfers across Caliciviridae / Picornaviridae at ≈ 0.59 (chance) vs. helicase L31 at ≈ 0.76 ($z = 8.5$ vs shuffle null).
- Linear probe trained to identify helicase generalized to held-out Picornaviridae samples (0.69 vs. k-mer 0.59).

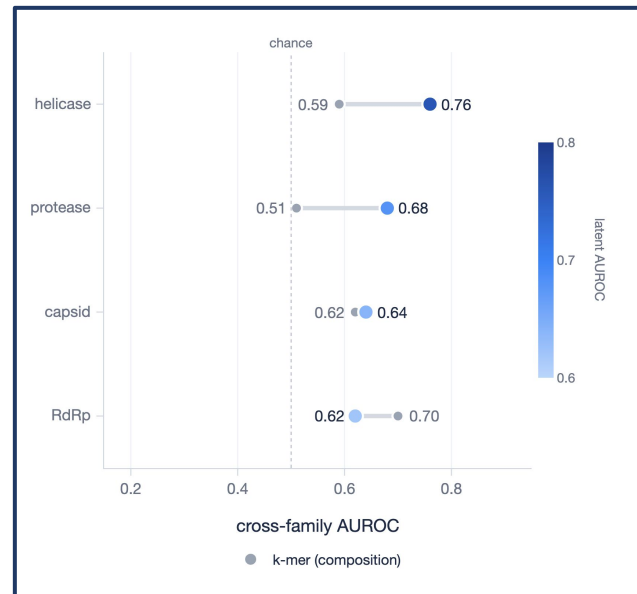


Fig. 8. CROSS-FAMILY FUNCTIONAL LATENTS. The best cross-family SAE latent per gene beats k-mer ladder baseline on transfer from Caliciviridae to Picornaviridae.

DISCUSSION: Conclusions

METAGENE LEARNS TO REPRESENT GENE FUNCTION INDEPENDENT OF ORGANISM.

- I. METAGENE-1 linearly represents pathogenicity (L31, 0.965 AUROC) and organism families (L24, 0.926 AUROC).
- II. Organism representations are primarily compositional, and subtracting reconstructed activations drops AUROC to 0.53.
- III. Sparse autoencoders trained on METAGENE-1 uncover gene-specific cross-family latents that outperform compositional baselines (0.761 vs 0.593 k-mer).

DISCUSSION: Limitations & Future

METAGENE-1 learns functional biological features in a non-compositional manner

Limitations:

- After deduplication and labeling, most of our labeled reads had very small sample sizes, which could be interfering with results.
- These experiments don't show that the latents are used by the model (no causation).
- Lack of knowledge of what biological information these latents fire for, or if they generalize to truly novel pathogens beyond homology lookup.

Future:

- Generate activations using the a larger wastewater dataset.
- Train SAEs and transcoders on all 32 layers of METAGENE to build an attribution graph and examine causality.
- Test on samples where homology lookup fails.

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